Lab Report 3: Bell-State Measurements

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Experimental Setup Used: #1

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1 Abstract

In the experiment, our group analyzed quantum entanglement through Bell-state measurements using spontaneous parametric down conversion (SPDC). A nonlinear crystal was pumped using a 405 nm laser at two different powers, 26.4mW and 3.16mW, to produce entangled pairs. Coincidence counts were recorded for sixteen different polarization angle combinations to test violations of the Bell inequality. Data was collected using five-second acquisition periods and processed through a spreadsheet to calculate S, the Bell parameter, including normalization and error analysis. The resulting S values violated the classical Bell inequality, demonstrating quantum entanglement.

2 Introduction

Quantum entanglement is a non-classical phenomenon where two particles remain correlated in a state where the measurement of one instantaneously determines the state of the other. This defies the classical concepts of locality. To examine whether such correlations are quantum or explained by variables, physicist John Bell found an inequality that sets a limit on classical theories.

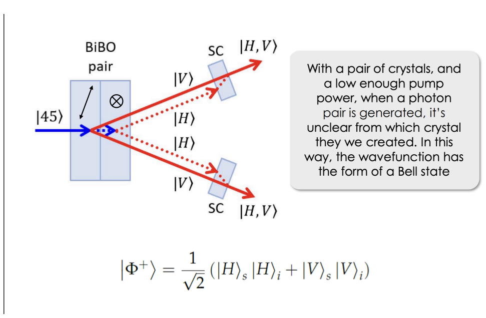

Figure 1: Diagram of entangled Bell States from lecture slides, p.15

The CHSH form of Bell's Inequality introduces parameter S, which is calculated from polarized angle measurements. If the magnitude of S ≥ 2, then classical theory is violated, showing quantum properties. This lab explores such violations.

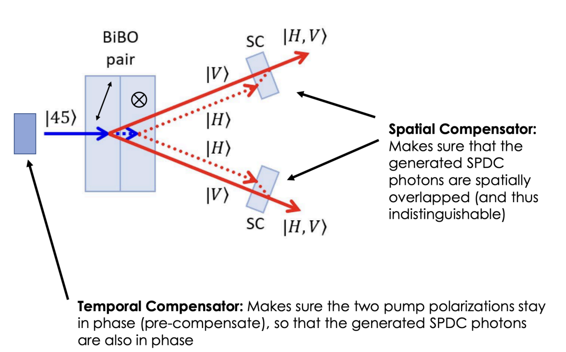

Figure 2: Entangled Pair Generation from SPDC form lecture slides, p.17

By measuring the number of coincidences at various angles, we can measure the correlation coefficients $E(\theta_s, \theta_i)$ (eq 2) which detects the likelihood of both photons being detected with the given polarization angles. The Bell parameter S (eq 1) can be defined as:

$$S = \left|E(\theta_s, \theta_i) - E(\theta_s, \theta_i')\right| + \left|E(\theta_s', \theta_i) + E(\theta_s', \theta_i')\right| \quad (1)$$

$$E(\theta_s, \theta_i) = cos[2(\theta_s - \theta_i)] \quad (2)$$

This experiment analyzes whether measuring polarization angle values violates Bell's inequality, reinforcing the quantum nature of the photon pair.

3 Procedure

This experiment was designed to measure entanglement using Bell-state analysis of SPDC-generated photon pairs. The setup includes a 405 nm pump laser directed through a pair of linear BiBO crystals. When the pump beam is set to a polarization of 22.5 degrees by the half-wave plate, both crystals contribute the same portion to the photon pair generation.

The idler photon is detected by the first of three single-photon avalanche diode (SPAD) detectors and serves as the herald. The signal photon's path is a longer optical path, and it is put through a 50:50 beam splitter to make sure it is a basic independent photon. This is then sent to two individual SPADs. The sixteen coincidence operations are measured by rotating the polarizers through the computer GUI.

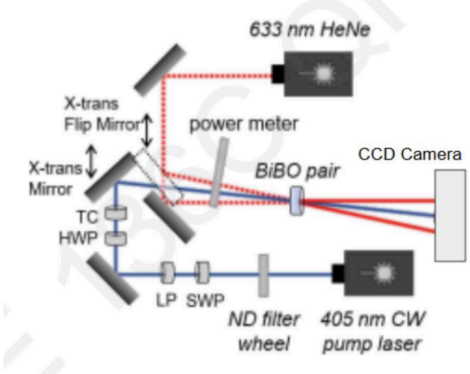

Figure 3: Experimental Setup

Photon detections (N) are recorded using the coincidence counting unit (CCU) in the setup. Using an acquisition time of 5 seconds allowed for a more accurate data set. Our goal is to use the measurements and determine the Bell state parameter S.

Using an initial pump power of ~25 mW, an initial count of accidental coincidences is measured by blocking the pump with the power meter to improve the accuracy of the data set. After removing the power meter, 16 different pairwise measurements were made using a combination of different angles for both signal and idler polarizers. After obtaining the coincidence count, the experiment was repeated, with a pump power of ~3 mW. Using the defined values for variable E and parameter S, we computed the results(as well as the error margin) to obtain an answer on whether the classical theory is violated

We used a spreadsheet to tabulate and calculate the four E values easily, and S to determine if we were able to observe a Bell-inequality violation. We also calculated ΔN and $\frac{\delta S}{\delta N}$ for each individual measurement in order to determine the error in our S parameter ΔS. This was done repeatedly for both power settings, with and without accidental coincidence subtraction.

4 Results and Discussion

4.1 Measurement 1

To measure our variable E, four pairwise measurements must be performed using combinations of different polarized signal and idler photons. Furthermore, the CHSH Bell-type inequality defines parameter S as:

$$S = |E(\theta_s, \theta_i) - E(\theta_s, \theta_i')| + |E(\theta_s', \theta_i) + E(\theta_s', \theta_i')|$$

Where θ_s and θ_s' are orthogonal angles for the signal photon and θ_i' and θ_i' are orthogonal angles for the idler photon. To obtain S, 16 pairwise measurements have to be made, as shown in Table 1.

θ_s	θ_i	$N(\theta_s, \theta_i)$
0	22.5	182
45	22.5	199
90	22.5	39
135	22.5	56
0	67.5	41
45	67.5	205
90	67.5	204
135	67.5	44
0	112.5	48
45	112.5	66
90	112.5	127
135	112.5	90
0	157.5	199
45	157.5	55
90	157.5	21
135	157.5	59

Table 1: Coincidence Counts for 26.4mW pump power

4.2 Measurement 2

To observe the dependency of the parameters on the pump power, the experiment is repeated using a power level of 3.16 mW. Although data acquisition time was recommended to be higher to improve signal to noise ratio, this decision was not taken into consideration.

θ_s	θ_i	$N(\theta_s, \theta_i)$
0	22.5	23
45	22.5	35
90	22.5	4
135	22.5	8
0	67.5	3
45	67.5	29
90	67.5	19
135	67.5	7
0	112.5	6
45	112.5	8
90	112.5	20
135	112.5	11
0	157.5	25
45	157.5	7
90	157.5	2
135	157.5	8

Table 2: Coincidence Counts for 3.16mW pump power

4.3 Measurement 3

Using the acquired coincidence counts, the following equations for parameters S and E can be applied:

$$E(\theta_s, \theta_i) = P(\theta_s, \theta_i) + P(\theta_s + \frac{\pi}{2}, \theta_i + \frac{\pi}{2}) - P(\theta_s, \theta_i + \frac{\pi}{2}) - P(\theta_s + \frac{\pi}{2}, \theta_i)$$

$$E(\theta_s, \theta_i) = \frac{N(\theta_s,\theta_i) + N(\theta_s + \frac{\pi}{2}, \theta_i + \frac{\pi}{2}) - N(\theta_s, \theta_i + \frac{\pi}{2}) - N(\theta_s + \frac{\pi}{2}, \theta_i)}{N(\theta_s,\theta_i) + N(\theta_s + \frac{\pi}{2}, \theta_i + \frac{\pi}{2}) + N(\theta_s, \theta_i + \frac{\pi}{2}) + N(\theta_s + \frac{\pi}{2}, \theta_i)}$$

$$S = |E(0, 22.5) - E(0, 67.5)| + |E(45, 22.5) - E(45, 67.5)|$$

The first equation shown determines our variable E using the probability of detecting a photon pair. Since probability is directly proportional to coincidence counts, the second equation is used to determine E. The second equation is normalized as the photon detection optics are not 100% efficient. Using the obtained correlation parameters, it can be applied to the third equation to determine if our parameter S.

Without taking accidental coincidence counts into consideration, the values for E are shown:

Experiment 1

$$E(0, 22.5) = 0.5606060606$$
$$E(45, 22.5) = 0.4063260341$$
$$E(0, 67.5) =- 0.7333333333$$
$$E(45, 67.5) = 0.4545454545$$
$$S = 2.154810883$$

Experiment 2

$$E(0,\ 22.5) = 0.6226415094$$
$$E(45,\ 22.5) = 0.4838709677$$
$$E(0, 67.5) =- 0.7959183673$$
$$E(45, 67.5) = 0.4509803922$$
$$S = 2.353411237$$

Since there is background noise due to other sources of light pollution in our system, we take our initial accidental counts and apply the following equation:

$$N_{new} = N_{old} - (N_s * N_i * \Delta T)$$

With this, Tables 3 and 4 are obtained, giving a more accurate look at the incident counts.

θ_s	θ_i	$Nnew(\theta_s, \theta_i)$
0	22.5	181.9983
45	22.5	198.9983
90	22.5	38.9983
135	22.5	55.9983
0	67.5	40.9983
45	67.5	204.9983
90	67.5	203.9983
135	67.5	43.9983
0	112.5	47.9983
45	112.5	65.9983
90	112.5	126.9983
135	112.5	89.9983
0	157.5	198.9983
45	157.5	54.9983
90	157.5	20.9983
135	157.5	58.9983

Table 3: Coincidence Counts After Accidental Subtraction (Pump Power = 26.4 mW)

θ_s	θ_i	$Nnew(\theta_s, \theta_i)$
0	22.5	22.9983
45	22.5	34.9983
90	22.5	3.9983
135	22.5	7.9983
0	67.5	2.9983
45	67.5	28.9983
90	67.5	18.9983
135	67.5	6.9983
0	112.5	5.9983
45	112.5	7.9983
90	112.5	19.9983
135	112.5	10.9983
0	157.5	24.9983
45	157.5	6.9983
90	157.5	1.9983
135	157.5	7.9983

Table 4: Coincidence Counts After Accidental Subtraction (Pump Power = 3.16mW)

Where N_{new} is the expected coincidence counts after accidental coincidences have been taken into account.

Using a more improved data set, the same previously stated equations can be applied to get more accurate measurements on whether or not the entangled pairs are violating local and realistic theories.

Experiment 1(without accidentals)

$$E(0, 22.5) = 0.5606154685$$
$$E(45, 22.5) = 0.406332604$$
$$E(0, 67.5) =- 0.7333438137$$
$$E(45, 67.5) = 0.454553776$$
$$S = 2.154845662$$

Experiment 2(without accidentals)

$$E(0,\ 22.5) = 0.6227195891$$
$$E(45,\ 22.5) = 0.4839228365$$
$$E(0, 67.5) =- 0.7960263248$$
$$E(45, 67.5) = 0.4510391634$$
$$S = 2.353707914$$

These measurements show a slightly larger S value, correlating to a stronger violation of Bell's Inequality.

4.4 Measurement 4

To quantify the uncertainty in our measured Bell parameter S, we used the following propagation of error formula:

$$\Delta S = \sqrt{\sum_{m=1}^{16} (\Delta N_m \frac{\delta S}{\delta N_m})^2}$$

Each correlation value E is calculated using four coincidence counts, and its uncertainty is derived using:

$E(\theta_\square, \theta_i) = ([N(\theta_\square, \theta_i) + N(\theta_\square + 90°, \theta_i + 90°) - N(\theta_\square, \theta_i + 90°) - N(\theta_\square + 90°, \theta_i)]) / ([N(\theta_\square, \theta_i) + N(\theta_\square + 90°, \theta_i + 90°) + N(\theta_\square, \theta_i + 90°) + N(\theta_\square + 90°, \theta_i)])$

The standard deviation ΔN in each count is assumed to follow Poisson Statistics, i.e. $\Delta N = \sqrt{N}$, and each partial derivative is computed analytically.
Using the measured coincidence counts at 26.4 mW, we determined:

$$E(0,\ 25)\ =\ 0.5606$$
$$E(0,\ 67.5)\ =\ -\ 0.7333$$
$$E(45,\ 22.5)\ =\ 0.4063$$
$$E(45, 67.5)\ =\ .4545$$

Thus, the calculated Bell Parameter is:
S=|E(0, 22.5) - E(0, 67.5)| + |E(45, 22.5)+ E(45, 67.5)| = 2.155

And propagated uncertainty in S is:
$\Delta S =.083$

This error estimate accounts for all statistical uncertainties in the coincidence count measurements used in calculating S. The final result of
$S\ =\ 2.155\ \pm\ 0.083$
indicates a statistically significant violation of the classical Bell inequality (S≤2), reinforcing the quantum-entangled nature of the photon pairs.

4.5 Discussion

Through the CHSH Bell state measurements, a non-classical view of photon entanglement was successfully measured. Using an initial higher pump power of ~25mW, the measured S parameter exceeded 2, clearly violating local realism and confirming the presence of quantum entanglement. For the higher power setting, the S parameter was measured to be $S\ =\ 2.15\ \pm\ 0.01$ indicating high precision with uncertainties resolved to the hundredth decimal place. Reducing the pump power to ~3mW showed similar non-classical behavior. The S parameter measured is $S\ =\ 2.35\ \pm\ 0.09$. Although this value still violates Bell's Inequality, the larger uncertainty reflects a smaller signal-to-noise ratio. This suggests that a higher data acquisition time should have been used as there are less photon pairs being measured. Other possible reasons for our uncertainties come from possible small equipment malfunctions, such as misalignments and/or photon detector inefficiencies.

Furthermore, data showed that accounting for accidental coincidence photon counts yielded higher S parameters further emphasizing the importance of our corrections in order to reveal stronger quantum correlations.

5 Conclusion

In this lab, we successfully measured Bell-state entanglement using photon pairs generated via SPDC. By performing 16 coincidence measurements across varying polarizer angles, we computed the CHSH Bell parameter S for two pump powers. Our results for both 26.4 mW and 3.16 mW pump powers yielded S values greater than 2, confirming a violation of the classical Bell inequality and indicating quantum entanglement.

We observed that subtracting accidental coincidences and using lower pump power both increased the measured S parameter. Lower pump power minimized background noise, improving entanglement accuracy. Our findings support the non-classical nature of the entangled photon pairs, reinforcing key principles of quantum mechanics and validating the effectiveness of Bell-state measurements in distinguishing quantum correlations from classical behavior.